

Week 1.10 — The Org Chart Goes to Work — A New Motor Driver, and the Second Splash

Sunday, August 23, 2026 · 3:00 – 5:00 PM · 13 students

Thirteen of us at 1.10 — Trenton, Isaac, Luke, Maddox, Carter, Rebecca, Sam, Frankie, Oliver, Eli, Rhylan, Tyler, and Adrian — from 3:00 to 5:00. This was the first session run with the structured roles we elected at 1.9, and the honest report is that it worked. People knew what they owned and went and did it instead of waiting to be told. The build headline is that the one-direction motor fault we have been carrying since 1.7 is finally closed. The electrical track swapped in the replacement Sabertooth motor driver, found and resoldered a loose wire while they were in there, and all four motors now run in both directions. Then we took the vehicle back to the water for the second pool session, and the second splash told us more than the first one did — because this time we knew what to look for. The nose sits up, the vehicle is slow going forward, it gets noticeably harder to move the farther out it goes as the tether drags, and it turns fast. Four observations, all of them useful, all of them written down. Meanwhile the mechanical track took the TriggerFish CAD model to roughly forty percent, the software track laid out the computing architecture it wants and started building the purchase case to take to the CFO, the sensor track ruled out analog video and named a digital camera candidate, and the payload track sketched claw concepts and ran into the real question — how the joints actually move. Leadership spent their time on the unglamorous half: surveying sub-roles, getting a schedule built before the game releases, and putting together a rough bill of materials.

4/4

MOTORS RUNNING AFTER
THE SABERTOOTH SWAP

2nd

POOL SESSION OF THE
SEASON

40%

TRIGGERFISH CAD MODEL
COMPLETE

13

STUDENTS AT THE SESSION

SECOND-SPLASH OBSERVATIONS

WHAT WORKED

- All four motors run in both directions. The replacement Sabertooth plus a resoldered joint closed the fault we had been carrying since 1.7.
- The vehicle turns fast — yaw authority is not a problem, which tells us the horizontal thrust is there even if the forward speed is not.
- The first session with real roles ran itself. People knew what they owned and worked without being assigned tasks.
- We went into this splash with a list of things to watch, so the observations are specific instead of "it felt slow."

WHAT NEEDS ADDRESSED

- The front of the ROV rides nose-up, every time. That is a trim problem, and it is almost certainly related to the next item.
- Forward speed is poor. A vehicle that flies nose-up is pushing water with its underside instead of moving through it — the two findings may be one finding.
- Tether drag scales with distance. The farther out the vehicle goes, the harder it is to move, which means the tether needs buoyancy management, not just more thrust.
- The control box is still analog. It got us into the water, but it cannot do what a mission run needs, and digitizing it is now an owned task.

DECISIONS MADE

- The Sabertooth swap closes the 1.7 motor fault. Replacement driver in, loose wire resoldered, all four motors verified in both directions before the vehicle went in the water.
- The analog control box becomes a digital one. Analog got the TriggerFish into the pool; it is not the path to a vehicle that can fly a mission, and the conversion is now on somebody's plate.
- The computing architecture is settled in principle: a Raspberry Pi as the central hub, an Arduino handling the real-time motor signals, and the Pi carrying data topside. The Pi talks to the Arduino, and the Arduino talks to the motors.
- Analog video is ruled out. The SeaMATE TriggerFish Barracuda kit works, but a digital camera is what serves the image-recognition and modeling goals we have set — so the extra effort is worth it.
- The Raspberry Pi Camera Module 3 Wide NoIR is the sensor track's candidate to beat.
- Hardware requests go through the CFO with a written justification. The Raspberry Pi 4 Model B and the Arduino Mega 2560 R3 both get reviewed and presented before any money moves.
- The season schedule gets built before the game releases in November — not after, when everyone is reacting.

WHAT WE LEARNED

- A motor that only runs one direction looks like a code problem right up until you swap the driver. We spent weeks with that fault; the fix was a new Sabertooth and a resoldered joint.
- Tether drag is a real force, not a nuisance. The vehicle got measurably harder to move the farther out it went, and no amount of extra thrust fixes a tether that is dragging — that is a buoyancy problem on the tether itself.
- Nose-up trim and slow forward speed are probably the same problem. A vehicle flying nose-high is shoving water with its belly. Chase the trim first and the speed may come with it.
- Choosing digital video over analog is a goals question, not a quality question. The Barracuda analog kit works fine — it just cannot feed the image recognition and modeling we want to do.

ACTION ITEMS

Survey sub-roles so every lead knows who is on their sub-team

All leads · by 1.11

Build the season schedule before the 2027 game manual drops in November

Annabelle (COO) · by 1.11

Turn the rough bill of materials into a costed list

Maddox (CFO) · by 1.11

Write the purchase case for the Raspberry Pi 4 Model B and Arduino Mega 2560 R3, present to CFO

Oliver (Software) · by 1.11

Diagnose the nose-up trim and the poor forward speed — treat them as one problem first

Frankie (Test) / Mechanical · by 1.11

Start converting the control box from analog to digital

Electrical group · by 1.11

Carry the TriggerFish CAD model past forty percent toward a complete assembly

Adrian (Mechanical) · by 1.11

Decide how the claw joints move — hydraulics or gears — and sketch the winner

Eli (Payload) · by 1.11

Work out how the motors and the microcontroller actually talk to each other, signal by signal

Oliver (Software) · by 1.11

MADE POSSIBLE BY



- The three-tier computing split has a reason behind it: the Pi is good at networking and video, the Arduino is good at precise timing, and asking either one to do the other's job is where teams lose their season.
- Wanting a part and getting a part are different steps. A purchase request goes to the CFO with a written case for why it is the right part — that is how a real company spends money.
- A claw is a joints problem before it is a claw problem. Hydraulics or gears is the decision that determines everything downstream.

STUDENT LEARNING OUTCOMES

- Ran the first working session under the elected org chart — leads owned their tracks without being assigned tasks.
- Replaced a failed motor driver and resoldered a bad joint, then verified all four motors in both directions before going in the water.
- Completed a second pool session and produced four specific, written handling observations instead of general impressions.
- Took a CAD model of the TriggerFish to roughly forty percent.
- Specified a computing architecture — Raspberry Pi as hub, Arduino for real-time motor control — and learned why the split exists.
- Compared analog and digital video paths and made a documented choice tied to team goals.
- Learned how a purchase request works inside a company: written justification, reviewed, presented to the CFO.
- Sketched manipulator concepts and identified the joint-actuation decision as the real design question.

LOOKING AHEAD

Class 1.11 moves to the desk. The mechanical track calculates buoyancy properly instead of guessing at foam, the electrical track learns to read schematics, the sensor track runs real comparisons on cameras and pressure sensors, and management works the integrated master schedule and the budget. The trim and speed findings from this splash go to the top of the mechanical list.